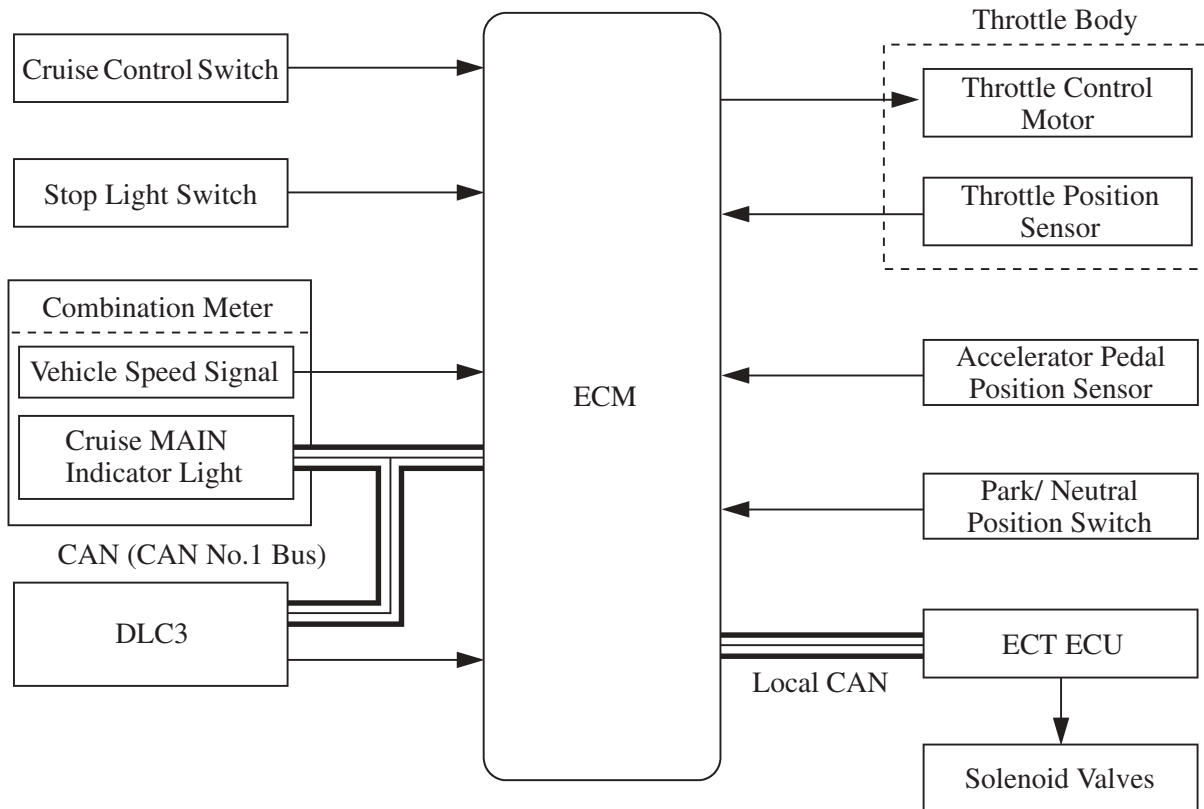


CRUISE CONTROL SYSTEM

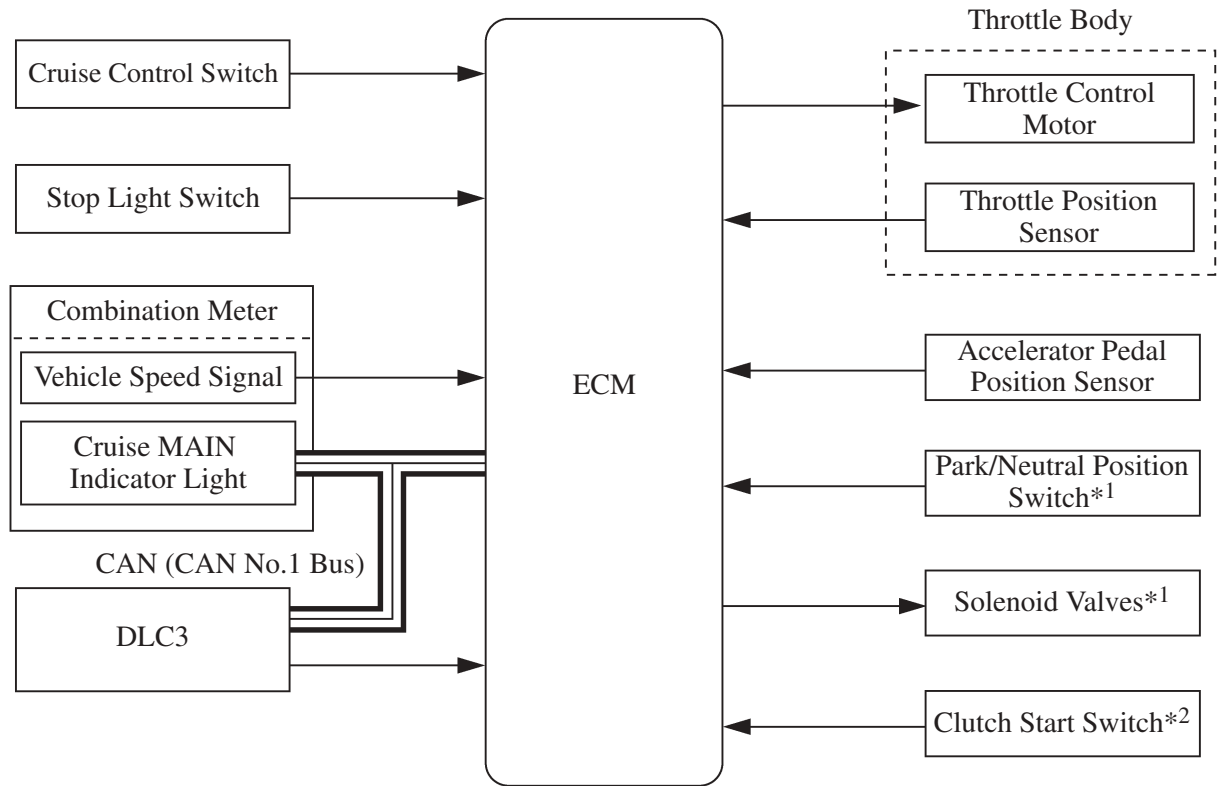
■ DESCRIPTION

- The cruise control system is optional equipment on all models.
- When the system is set to a desired vehicle speed, the throttle valve position is adjusted automatically to maintain the vehicle speed without the driver having to depress the accelerator pedal.
- This system effects control through the ETCS-i (Electronic Throttle Control System-intelligent).

► Models With U660E Automatic Transaxle ◀



► Models With U250E Automatic Transaxle and E351 Manual Transaxle ◀

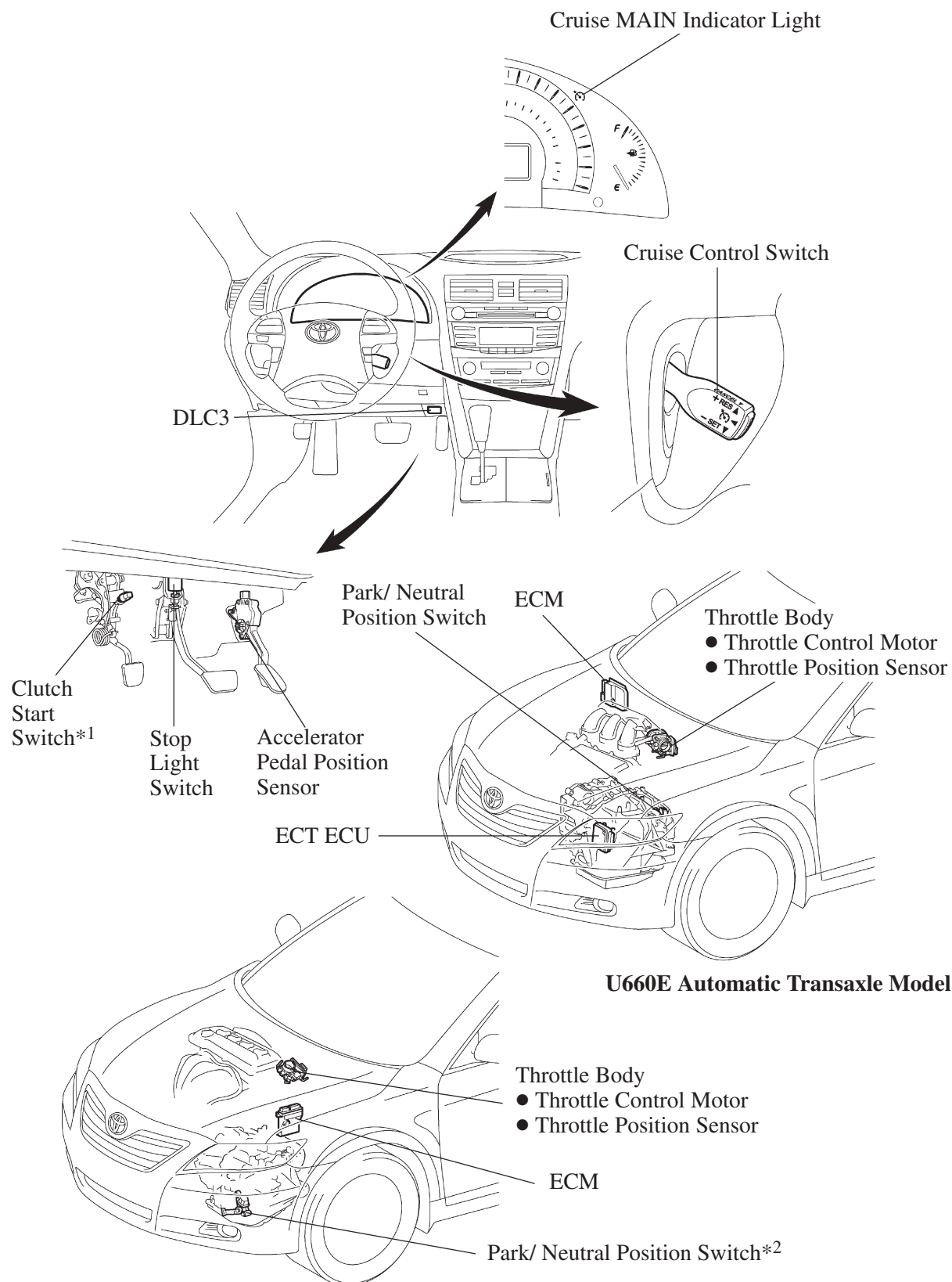


025BE117P

*1: Only for U250E automatic transaxle

*2: Only for E351 manual transaxle

2. Layout of Main Components



025BE118TE

*1: Only for E351 Manual Transaxle Model

*2: Only for U250E Automatic Transaxle Model

3. System Control

General

The cruise control has the following control.

Control	Outline
Constant Speed Control	The ECM compares the actual vehicle speed and the set speed and if the vehicle speed is higher than the set speed, it uses the throttle control motor to decrease the throttle opening. If the actual vehicle speed is lower than the set speed, it uses the throttle control motor to increase the throttle opening.
Set Control	While this system fulfils the following conditions, and the cruise control switch is pressed to the SET/– side and released when the ON-OFF button on the cruise control switch has been pressed to turn the system on, the ECM stores the vehicle speed and maintains the vehicle constantly at that speed. ● The vehicle is running at a vehicle speed of about 40 km/h (25 mph) or more.
Low Speed Limit Control	The low speed limit is the lowest speed that cruise control can be set at and it is designed to be approx. 40 km/h (25 mph). The cruise control cannot be set below that speed. If the vehicle speed drops below that speed while running in the cruise control, the cruise control will be cancelled automatically. However the set speed in the memory is kept.
COAST Switch Control	While the cruise control switch is held to the SET/– side, the vehicle speed and the set vehicle speed change as follows. ● The vehicle decelerates constantly. ● The set vehicle speed changes to the speed that the vehicle is traveling at when the COAST switch is released.
Tap Down Control	When the cruise control switch is pushed momentarily (approx. 0.6 sec.) to the SET/– side, the vehicle speed and the vehicle setting speed change as follows. ● The vehicle will decelerate in increments of approx. 1.6 km/h (1 mph) for each time the switch was pressed. ● However, if the difference between the actual vehicle speed and the vehicle setting speed is greater than 5 km/h (3.1 mph), the vehicle setting speed will change to the speed at which the vehicle was being driven at the time the switch was operated.
ACC Switch Control	When the cruise control switch is pushed to the RES/+ side and held, the vehicle speed and the vehicle setting speed change as follows. ● The vehicle accelerates constantly. ● The set vehicle speed changes to the speed as which the switch is releases.
Tap Up Control	When the cruise control switch is pushed momentarily (approx. 0.6 sec.) to the RES/+ side, the vehicle speed and the vehicle setting speed change as follows. ● The vehicle accelerates in increments of approx. 1.6 km/h (1 mph) for each time the switch was pressed. ● However, if the difference between the actual vehicle speed and the vehicle setting speed is greater than 5 km/h (3.1 mph), the vehicle setting speed does not change.
RES Switch Control	If cruise control is canceled for any reason other than a malfunction or main switch operation and vehicle speed is more than the low speed limit, the vehicle speed is returned to the speed before the cancellation of cruise operation by setting the cruise control switch to the RES/+ side. The cruise control mode can be resumed even if the vehicle speed drops below the low speed limit, because the speed in the memory is not cleared.

(Continued)

Control	Outline
Shift Down Control	ECT ECU incorporated into ECM (U250E Automatic Transaxle): When the vehicle is cruising uphill, shift-down control may be performed by the ECT (Electronic Control Transmission). When the ECM judges the end of cruising uphill based on the throttle valve angle, the shift-down control will turn on again. There is a case where the shift-down control turns off during ACC or RES switch control.
	ECT ECU isolated from ECM (U660E Automatic Transaxle): When the vehicle is cruising uphill, shift-down control may be performed by the ECT (Electronic Control Transmission). The ECM transmits the shift up request signal to the ECT ECU when the ECM judges the end of cruising uphill based on the throttle valve angle. If shift-down control is performed during ACC or RES switch control, the ECM transmits the shift up request signal to the ECT ECU after ACC or RES switch control is completed.
Manual Cancel Control	If any of the following signals is sent to the ECM, the cruise control is cancelled accordingly. ● Clutch switch ON signal/Depress the clutch pedal. (for M/T) ● Stop light switch ON signal/Depress the brake pedal. (for A/T) ● CANCEL switch ON signal (cruise control switch moved to CANCEL side) ● Cruise control switch (ON-OFF button) OFF signal.
Automatic Cancel Control	When any of the following conditions occur during cruise control operation, the speed that is set in the memory is cleared and the cruise control is cancelled. ● Stop light switch open or short circuit ● The vehicle speed signal is not input for a predetermined period of time. ● ETCS-i malfunction Furthermore, the cruise MAIN indicator light will blink until the ON-OFF button on the cruise control switch is used to turn the system off, and the operation of the cruise control will be disabled until the ON-OFF button is turned ON again.
	When any of the following conditions occur during cruise control driving, the speed that is set in the memory is cleared and the cruise control is cancelled. ● Stop light switch input signal is abnormal. ● Cruise control switch input signal is abnormal. Furthermore, the cruise MAIN indicator light will blink until the ON-OFF button on the cruise control switch is used to turn the system off, and the operation of the cruise control will be disabled until the power source* is turned IG-ON again.
	When any of the following conditions occur during cruise control driving, the cruise control is cancelled. ● Vehicle speed is below the low speed limit (approx. 40 km/h [25 mph]) or less. ● Vehicle speed decreases by 16 km/h (10 mph) or more below the speed at which the cruise control was set.
Diagnosis	When the ECM does not receive a vehicle speed signal for a predetermined period of time during cruising, or when cruise control is cancelled (automatic cancel) due to a malfunction of the cruise control, stop light switch or vehicle speed signal, the ECM immediately blinks the cruise MAIN indicator light due to the malfunction. The contents relating to the malfunction will be stored in the ECM.

*: Power source conditions can be changed by pressing the engine switch on models with the smart key system and by operating the ignition switch on models without smart key system.

Diagnosis

If a malfunction occurs in the cruise control system, during cruise control operation, the ECM actuates the automatic cancel control and blinks the cruise MAIN indicator light to inform the driver of a malfunction. At this time, the ECM memorizes the malfunction in the form of 5-digit and 2-digit DTC (Diagnostic Trouble Code).

- The 5-digit DTC can be read by connecting a hand-held tester to the DLC3.
- The 2-digit DTC is output to the cruise MAIN indicator light when the Tc and CG terminals of the DLC3 connector are connected through the use of the SST (09843-18040). Thus, these DTC can be obtained by counting the number of blinks of the cruise MAIN indicator light.

Service Tip

When using a hand-held tester, a dedicated adapter [CAN VIM (Vehicle Interface Module)] must be connected between the DLC3 and the hand-held tester. For details, see the 2007 Camry Repair Manual (Pub. No. RM0250U).